

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0243

Ross 69 kV
January 16, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0243. The Interconnection Customer proposes a 30.6 MW solar photovoltaic facility interconnecting at the Ross 69 kV substation in the North Carolina area. The request was received on June 02, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 165 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$272,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The E Sparta 138/69 kV Transformer Bank is overloaded to 102.0% of its 99 MVA emergency rating for the loss of the Roanoke Rapids / Edgecombe Genco Llc 345kV. The project contributes 5.1% DFAX (1.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Sparta 138/69kV TX	Replace transformer with higher-rated unit	\$4,074,000	7% (\$272,000)
	Total allocated to project		\$272,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.